5. Bernoulli's Equations (Reducible to Linear)

Bernoulli's equations are a specific class of **non-linear** differential equations that can be easily transformed into a standard **Linear** differential equation using a clever substitution. This method is the final standard technique for first-order ODEs.

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1. Concept Overview

A Bernoulli differential equation has the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Where n is any real number except 0 or 1. * If $n = 0$, the equation is Linear. * If $n = 1$, the equation is Variable Separable. * For any other n, the term y^n on the right side makes the equation **non-linear**.

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2. Working Rule (Reduction to Linear Form)

To solve a Bernoulli equation, we must "linearize" it.

Step 1: Divide by the non-linear term Divide the entire equation by y^n to move the y terms to the left side:

$$y^{-n}\frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

Step 2: Substitute Let $z = y^{1-n}$ (the term attached to $P(x)$).

Step 3: Differentiate Differentiate z with respect to x:

$$\frac{dz}{dx} = (1-n)y^{-n}\frac{dy}{dx}$$

Rearrange this to match the first term of your divided equation:

$$\frac{1}{1-n}\frac{dz}{dx} = y^{-n}\frac{dy}{dx}$$

Step 4: Transform into Linear DE Substitute z and $\frac{dz}{dx}$ back into the equation. It will now be a standard Linear DE in terms of z and x:

$$\frac{dz}{dx} + (1-n)P(x)z = (1-n)Q(x)$$

Step 5: Solve and Resubstitute Solve this new linear equation for z using the Integrating Factor method (see Topic 4). Finally, replace z with y^{1-n} to get the general solution.

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3. Solved Questions

Here are two step-by-step examples from the lecture notes.

Question 1: Rational Powers

Solve: $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

Solution:

Step 1: Identify Form This matches Bernoulli's form with $P(x) = 1/x$, $Q(x) = 1/x^2$, and $n = 2$.

Step 2: Divide by y^n (y^2)

$$y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = \frac{1}{x^2}$$

Step 3: Substitute Let $z = y^{1-2} = y^{-1}$. Differentiate: $\frac{dz}{dx} = -1 \cdot y^{-2}\frac{dy}{dx} \implies -\frac{dz}{dx} = y^{-2}\frac{dy}{dx}$.

Step 4: Transform Substitute into the equation:

$$-\frac{dz}{dx} + \frac{1}{x}z = \frac{1}{x^2}$$

Multiply by -1 to standardize:

$$\frac{dz}{dx} - \frac{1}{x}z = -\frac{1}{x^2}$$

Step 5: Solve Linear DE * **I.F.:** $e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$ * **Solution Formula:**

$$z \cdot \left(\frac{1}{x}\right) = \int \left[-\frac{1}{x^2} \cdot \frac{1}{x}\right] dx + C$$

$$\frac{z}{x} = -\int x^{-3} dx + C$$

$$\frac{z}{x} = -\left(\frac{x^{-2}}{-2}\right) + C = \frac{1}{2x^2} + C$$

Step 6: Resubstitute ($z = 1/y$)

$$\frac{1}{xy} = \frac{1}{2x^2} + C$$

Question 2: Cubic Power

Solve: $\frac{dy}{dx} + 4xy = xy^3$

Solution:

Step 1: Identify Form This is Bernoulli's form with $n = 3$.

Step 2: Divide by y^3

$$y^{-3}\frac{dy}{dx} + 4xy^{-2} = x$$

Step 3: Substitute Let $z = y^{1-3} = y^{-2}$. Differentiate: $\frac{dz}{dx} = -2y^{-3}\frac{dy}{dx} \implies -\frac{1}{2}\frac{dz}{dx} = y^{-3}\frac{dy}{dx}$.

Step 4: Transform Substitute into the equation:

$$-\frac{1}{2}\frac{dz}{dx} + 4xz = x$$

Multiply by -2 to standardize:

$$\frac{dz}{dx} - 8xz = -2x$$

Step 5: Solve Linear DE * **Identify P(x):** $P(x) = -8x$ * **I.F.:** $e^{\int -8x dx} = e^{-4x^2}$ * **Solution Formula:**

$$z \cdot e^{-4x^2} = \int (-2x)e^{-4x^2} dx + C$$

Step 6: Integrate To solve $\int -2xe^{-4x^2} dx$, use substitution $u = -4x^2 \implies du = -8xdx \implies \frac{1}{4}du = -2xdx$.

$$\text{Right Side} = \int e^u(\frac{1}{4}du) = \frac{1}{4}e^u = \frac{1}{4}e^{-4x^2}$$

So the equation becomes:

$$ze^{-4x^2} = \frac{1}{4}e^{-4x^2} + C$$

Step 7: Resubstitute ($z = y^{-2}$)

$$y^{-2}e^{-4x^2} = \frac{1}{4}e^{-4x^2} + C$$

Multiply by e^{4x^2}:

$$\frac{1}{y^2} = \frac{1}{4} + Ce^{4x^2}$$

Next Module: Now that we have mastered all methods for solving first-order equations, we will explore their geometric applications. Proceed to Module 3: Applications of First-Order ODEs.